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B2 Noise Figure

Noise factor F measures the degradation in signal to noise ratio caused by the system receive chain, compared to an ideal receiver; noise figure NF is simply noise factor expressed in dB.

$$F = \frac{SNR_{in}}{SNR_{out}} \quad \text{and} \quad NF = 10\log F \tag{D.1}$$

SNR_{in} is the signal to noise available at the system's input (generally taken as the preamplifier input) and SNR_{out} is at the system output (generally taken as an image). A system with F=1 (a 0dB noise figure) would be a perfect system; there would be no degradation in signal to noise anywhere in the system. A system with F=2 (a 3dB noise figure) has lost half the available SNR, due to imperfections in the system.

Noise factor and figure are relatively recent concepts; they date from the 1940s when Harold Friis created the definition above. Historically, "noise figure" was used for both linear and dB expressions. Modern usage is "noise factor" for the linear expression, and "noise figure" for the dB version, to avoid the obvious confusion.

If the system gain G is known, the equation above reduces to

$$F = \frac{SNR_{in}}{SNR_{out}} = \frac{S_i/N_i}{S_o/N_o} = \frac{S_i/N_i}{S_iG/N_o} = \frac{N_o}{GN_i} \tag{D.2}$$

Note that the output noise N_o is *not* equal to the "gained up" input noise N_iG, as the system always adds additional noise.

This relationship is generally impractical to measure, as it requires very accurate measurements of gain G and input noise N_i. But it does make clear that noise figure is just the ratio between the system's actual output noise N_o and what would be expected based on input noise N_i and system gain G.

Measuring F based on these definitions is not very practical, as determining SNR_{in} is hard; the signals are tiny, and there is no practical way to quantify noise at this level. Several practical alternatives exist.

B2.1 The *Y*-factor method

One practical alternative is called the Y-factor method. A noise source, capable of switching between two calibrated noise levels (typically called "on" and "off", and related by an *excess noise ratio*, or *ENR*), is fed into the system input. The output noise of the system is measured with the noise source at both levels. From the output noise measurements and *ENR*, the noise factor can be found.

The *Y*-factor is defined as the ratio of noise *power* at the system output, for the noise source on (N_2) and off (N_1). For an image, the noise power is the variance in pixels of a noise image. To get the variance, square the standard deviation of an ROI obtained from the noise on and noise off images.

$$Y = \frac{N_2}{N_1} = \left(\frac{\sigma_2}{\sigma_1}\right)^2 \tag{D.3}$$

The noise factor of the system is based on Y and ENR of the noise source:

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$$F = \frac{ENR}{Y-1} \qquad (D.4)$$

Example: A noise source with *ENR* of 5.33dB is used; the "noise on" and "noise off" images had ROI standard deviations of 7.23 and 4.10, respectively. The *Y*-factor would be

$$Y = \frac{N_2}{N_1} = \left(\frac{7.23}{4.10}\right)^2 \approx 3.11$$

ENR must be in the linear domain (not dB). Doing the conversion and calculating *F* and, in turn, *NF:*

$$F = \frac{10^{5.33/10}}{3.11-1} = \frac{3.41}{3.11-1} = 1.64, \text{ so } NF = 10\log(1.64) = 2.14 \text{ dB}$$

B2.2 Noise sources

One type of appropriate, commercially available noise source is based on avalanche noise diodes (such as the Agilent 346A/B.) Typically, versions are available with excess noise ratios of 5dB (ENR≈3.16) and 15dB (ENR≈31.6). A 5dB source is useful for measuring *NF* between 0 and about 5dB; if the expected *NF* is higher, a 15dB source should be used. A calibration table of *ENR* vs. frequency is supplied with the source, and should be used.

Using these sources is straightforward. A 28VDC supply switches between "on" and "off". Accurate measurements of low noise performance typically require shielding of the test setup from environmental noise.

In 2010, a 5dB source was around $3000.

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B3 Spherical Harmonics

MR uses spherical harmonics to work with magnets and gradients, as these harmonics provide an efficient way of accurately describing the magnet fields produced by these components. Spherical harmonics are used in magnet and gradient design analysis, magnet shimming, gradient spatial corrections (*gradwarp*), and other areas. They are also widely used in other specialties, including gravitational fields, geospatial systems, optics, and 3D computer graphics.

Spherical harmonics provide a complete set of orthogonal functions, each of which is defined over the surface of a sphere. The mathematics of spherical harmonics are covered in detail elsewhere (the link above is a good start) but an overview is provided here.

Figure B3-1 shows the shape, or spatial dependence, for the first six degrees of harmonics – the degrees shown are $l = 0 \ldots 5$, where each degree is one row in the figure. Each row has m orders, where $m = -l \ldots l$. Each harmonic, Y_l^m, has a specific spatial dependence. Since spherical harmonics form a complete set of basis functions, any function on the sphere's surface can be described by a linear sum of spherical harmonics (although more than the first six degrees may be needed!)

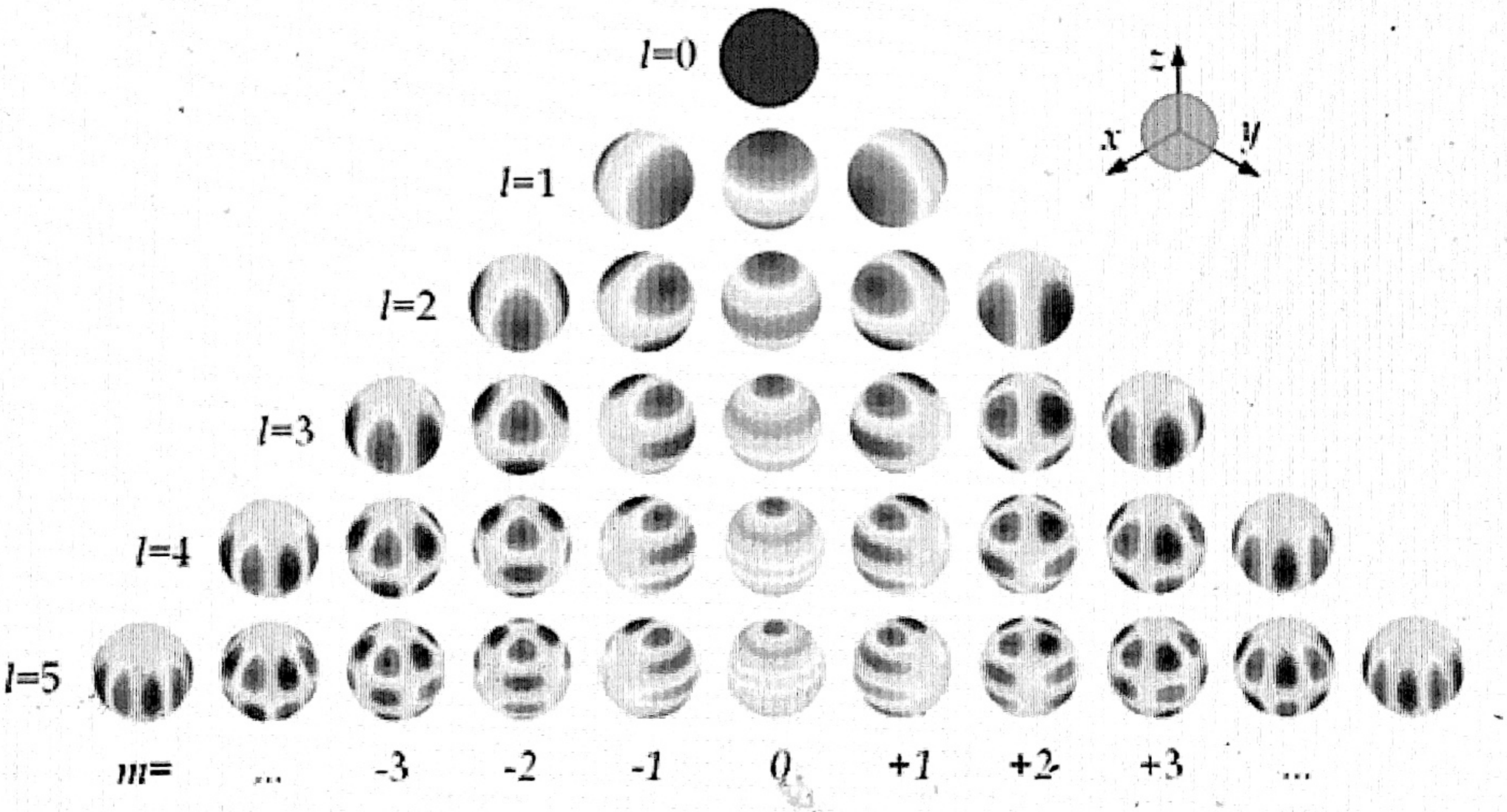


Fig. B3-1: Spherical Harmonics

In the figure, the colors represent the relative value on the surface of the sphere; deep red is +1.0, blue is -1.0. The Y_0^0 term is a constant; it has the same value everywhere on the sphere. Y_1^0 is a z term, where the value is proportional to the z-location. Y_1^1 is the x-term, and Y_1^{-1} is the y-term.

Note: GEHC cylindrical MRI systems always align the spherical harmonic z-axis with the magnet's z-axis; however, alignment of the x and y axes varies across GEHC MR. DOC1000959, *MR750w LVshim Tool SWDD*, Table 3: Harmonics transformation from Florence to Milwaukee Convention, may be helpful.

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Since any function can be expressed as a sum of spherical harmonics, these spherical harmonics are useful for characterizing MRI fields. For example, the B0 magnetic field strength on the surface of a sphere would have a dominant Y_0^0 term (since it is a homogeneous field), and much smaller higher order terms expressing the inhomogeneity. An x-gradient field would have a dominant Y_1^1 term, plus other smaller terms that express the gradient field's non-linearity.

In regions containing no currents (like inside the patient bore), knowing the spherical harmonics on the surface of a sphere allows reconstruction of the field anywhere *inside* the sphere as well. This property is useful for calculating shim at smaller volumes from measurements made on a larger sphere.

Another common visualization shows the harmonics as shapes, with the radius set to the value of the harmonic in a given direction. In Figure B3-2, light gray represents positive values, dark gray negative values, and z is where the surface is pulled to the center (i.e., r=0.)

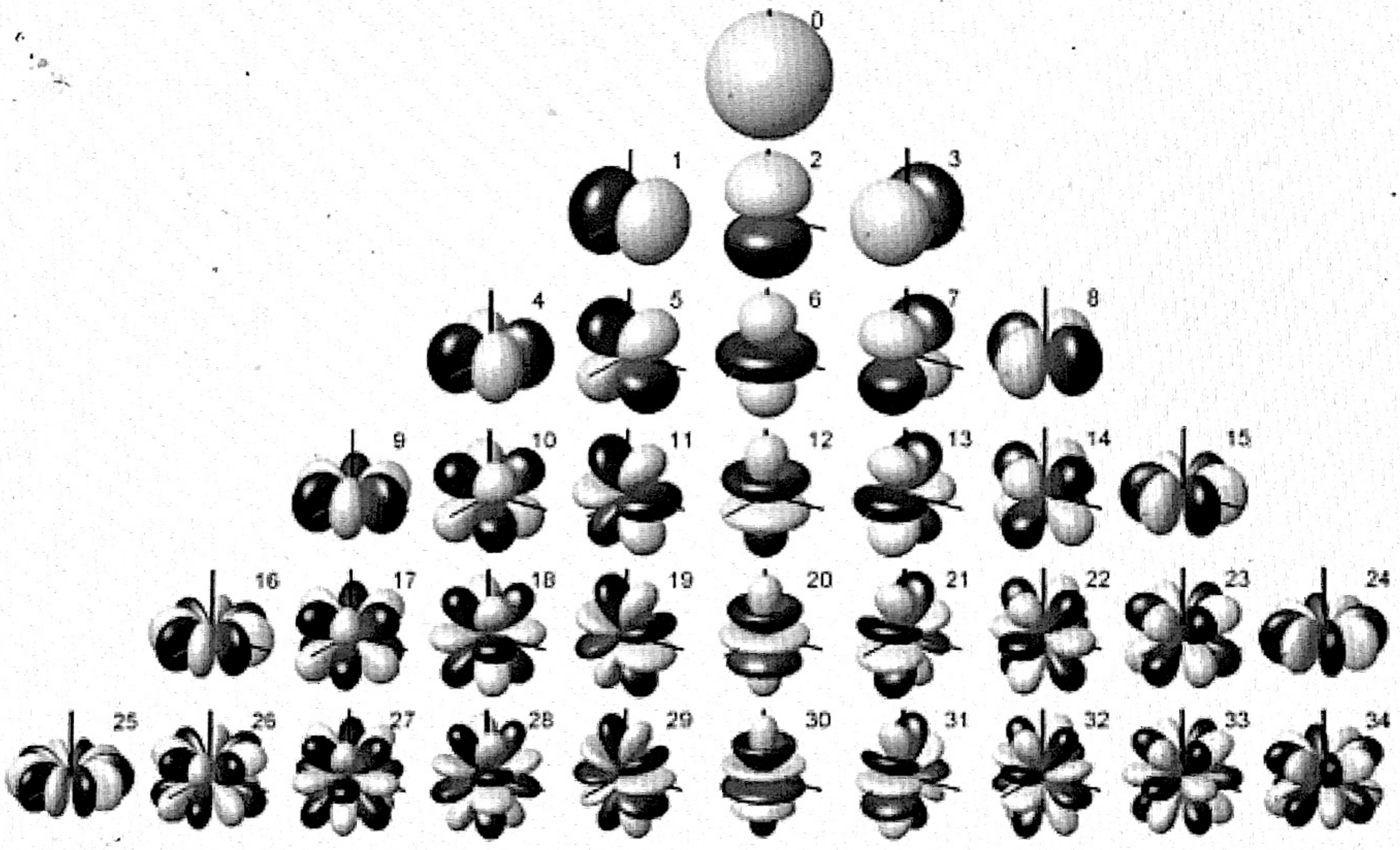


Fig. B3-2: Spherical harmonics seen as shapes

In this figure, the labelling numbers are arbitrary, but the position for each harmonic matches the first figure.

Despite their name, spherical harmonics can be expressed in Cartesian coordinates (x, y, z), or in spherical coordinates (θ, φ).

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Appendix C Maintaining the SDD

This section gives an overview of editing and maintaining the SDD document.

C1 Figures

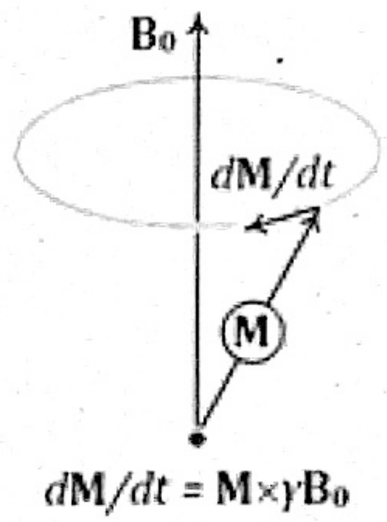
Most figures in this document originate as either 1) Visio drawings or GIF bitmaps of the drawing, or 2) images or photos.

Visio drawings

Visio drawings should (if practical) be drawn to be legible in Visio at 100% zoom. This allows they to be placed directly in Word as VSD (*.vsd) objects. In this case, they can be opened for editing with a right click.

Visio drawing may also be exported from Visio into a common graphics format, like GIF, PNG, or JPG, and this new file placed into the doc. GIFs are especially efficient for line drawings; JPGs tend to be better for picture and images.

For drawings, exporting as GIFs usually gives better display quality and smaller added filesize. The precession graphic at right is a GIF, exported from Visio at 300 DPI.



Since Word's display is scaled to 96 DPI, the graphic must be scaled to display at the correct size. After inserting the graphic, it is sized to 32% -- which is 96/300 = 0.32 or 32%.

Since the original resolution is 300 DPI, the graphic has decent quality even when zoomed in.

Images and photos

Images and photos add clarity and interest, but care is needed to avoid large file sizes. By default, the SDD is set to *not* compress or downsample images when they are imported. (This can be changed thru File→Options→Advanced settings.)

Word defaults to a 72 DPI display resolution, but display quality will improve if the resolution is two to four times that – around 140-300 pixels per inch.

The "How MRI Works" image below is a useful example to explore image size, scaling, and final display size. The original graphic is 400 x 400 pixels.

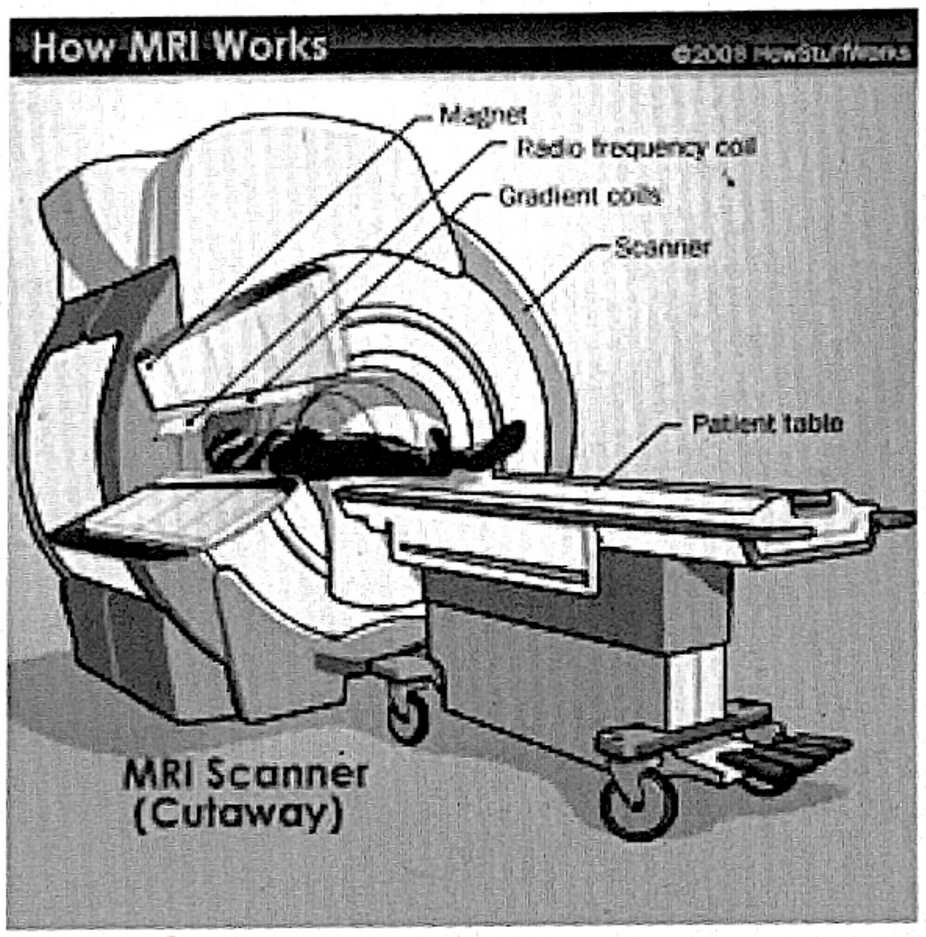
Choosing a display size of 3x3 inches gives 133 pixels per inch; on the low side, but still reasonable.

When imported, Word will set the image size to 5.56":

400 pixels / 72DPI = 5.56"

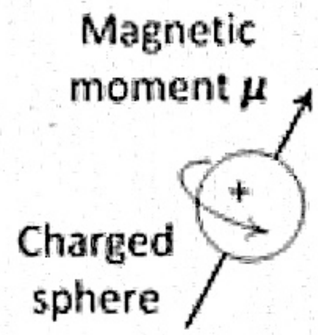
Setting the image scaling to 54% (or 3"/5.56") results in the image shown at right, which displays at the planned 3" size.

In general, choose the display size first. Multiply by 200 to estimate a reasonable resolution for image to be imported. A 2" display might need a 400 pixel image; a 5" display size might need 1000 pixels.



C2 Figure captioning

Each figure should be captioned, to describe the purpose and allow referring to the graphic. An example is below. The two-cell table provides a handy way to keep diagrams tied to text, but is not needed. Without the table, the graphic will be centered.

To add captions to figures, use two adjacent paragraphs. (Keeping two separate paragraphs makes adding cross-references much cleaner.) The upper paragraph has the actual graphic, and uses style "Graphic". The lower one has the caption, which usually consists of the figure number and title, and uses style "Graphic". Cut-and-paste both graphic and caption to set up a new figure; after pasting, select the caption and hit F9 to update the figure number.	 **Fig. 6–1: A graphic**

Note -- If you are saving the document in *.docx format, the figures will not display in Draft view. This is due to the way Word separates content into layers in the *.docx format.

C3 Math typography

Math notation in this document use these basic conventions:

- Single-letter symbols for variables, physical quantities, and physical constants are set in italic type. Examples: volume V, speed of light c, gyromagnetic ratio γ.
- Vectors are set in bold roman type. Examples: location $\mathbf{r}$, magnetization $\mathbf{M}$, static field $\mathbf{B}_0$
- Functions, numerals, operators, and fences are set in roman type. Examples: sin x, the difference operator Δ.

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- Functions set in roman type are bordered by a space – example $\sin\theta$ – unless the argument is enclosed in fences – example $\cos(3x)$.
- Superscripts and subscripts that are themselves symbols for variables or physical quantities are set in italic type, whereas superscripts and subscripts that are numerals or abbreviations are set in roman type: compare, for example, C_p for heat capacity at constant pressure and x_0 for an offset.
- Mathematical operators functioning as verbs or conjunctions are set with a space on each side (e.g., $n = 25$ and 6.022×10^{23}), except when they are used in subscripts or superscripts (e.g., a_{n+1}). One exception is the slash (/), which is usually set without spaces: $x/(y-z)$.
- Mathematical symbols used as adjectives are set without a space between the symbol and the number: for example, a v of <10 meaning a velocity of less than 10.

C4 Equations

Most of this document's math notation (variables, equations, etc.) uses MS Word's math notation support. The keyboard shortcut Alt+"=" creates an equation. (Some older sections may have equations created with MathType, or MS Equation Editor 3.0).

Inline vs. display formatting

Word's math notation support *inline* and *display* formats. Word automatically chooses between inline and display based on the context. Adding content into an existing paragraph uses inline. Placing an equation into an empty paragraph uses display formatting.

The most common use for inline format is to add content into a sentence, like "adjust Δx to be zero."

Display format is used for equations shown separately, like the quadratic equation example below.

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Equation layout and numbering

In the SDD, numbered equations are set into a table with two cells. The left cell has the equation; the right cell has the equation number. Both cells should use style "Equation". In the example below, the cell borders are turned on to show the cells, but the borders should be turned off for general use.

$\frac{d\mathbf{M}}{dt} = \mathbf{M} \times \gamma\mathbf{B} - \frac{M_x\mathbf{i} - M_y\mathbf{j}}{T_2} - \frac{(M_z - M_0)\mathbf{k}}{T_1}$	(6.7)

Equation numbers use the format "(3.12)" indicating the 12th equation in section 3. The numbering in the example above has field codes to automatically generates the equation number – after cut-n-paste, select the equation number and hit F9 to update the equation number.

C5 Math characters in "Cambria Math"

For the interested reader.

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Microsoft Word's math notation support uses the "Cambria Math" font by default. Cambria Math is a Unicode font, with about 4400 characters, and was the first font to implement the OpenType math extension.

Word's math support, including the equation editor, uses *alphanumeric math characters* to give better type control and legibility. Generally, the AN math characters are different than the standard characters. For example, the alphanumeric math character x (Unicode 1D465 MATHEMATICAL ITALIC SMALL X) is different from ASCII 78, the standard character for x, or in italics x. (The AN math characters start at Unicode 1D400 and end at 1D800.) Some AN math characters commonly used in the SDD: $x, y, z, \pi, \alpha, \beta, \gamma$

Using the math characters make the SDD easier to read and understand.

Using inline equations is an easy way to get the characters right. It is fairly efficient: to place the math character x inline, use Alt+"+" to create an inline equation, type "x", then right arrow the exit the equation.

As an alternative, use Word's Insert→Symbol, set font to Cambria Math, and scroll down to "Extended characters – Plane 1".